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Javascript fundamentals Assessment Tes

Topics: Variables & Data types, Operators & Expressions, Control flow

Section - A
Debug the Code

1. Find and fix the errors.

```
const name = "Luvin";
name = " Rahul";
console. log (name);
```

Answer:

```
let name = "Luvin";
name = " Rahul";
consolo. log ( name);
```

2. This Code throws an error. fix it,

```
console. log (age);
let age = 20;
```

Answer:

```
let age = 20;
console.log(age);
```

3. Find the mistake.

```
const user = {name: "John"};
user = {name: "David"};
```

Answer:

```
const user = {name: "John"};
user.name = "David";
console.log(user);
```

4. Fix the scope issue.

```
if (true) {
  let message = "Hello";
}
console.log(message);
```

Answer:

```
let message;
if (true){
   message = "Hello";
}
console.log(message);
```

5. Debug this loop.

```
for (let i=1; i<=5; i--){
  console.log(i);
}
```

Answer:

```
for (let i = 1; i<5; i++){
   console.log(i);
}
```

6. find the bug.

```
const arr = [1,2,3];
for (let value in arr){
console.log(value);
}
```

Answer:

```
const arr = [1, 2, 3];
for (let value of arr) {
    console.log(value);
}
```

7. fix the Comparison:

```
let Value = "10";
if (value === 10){
Console.log("Equal");
}
```

Answer:

```
let Value = "10";
if (value == 10) {
    console.log ("Equal");
}
```

8. Debug Optional chaining.

```
const User = null;
console.log( user?.address.city);
```

Answer:

```
const User = null;
Console.log (user?.address?.city);
```

9. Fix the Switch statement.

```
let day = 2;
switch (day) {
case 1: console.log ("monday");
case 2: console.log ("Tuesday");
default : console.log (" invalid");
}
```

Answer:

```
let day = 2;
switch (day) {
    case 1;
        console.log (" Monday");
        break;
    case 2;
        console.log ("Tuesday");
        break;
    default:
        console.log (" invalid");
}
```

10. Find the infinite loop.

```
let i = 1;
while (i <= 10) {
console.log(i);
}
```

Answer:

```
let i = 1;
while (i <= 10) {
    console.log(i);
    i++;
}
```

Section B

Predict the Output

11. Console.log (typeOf null);

Answer: Object

12. Console.log (type Of []);

Answer: object

13. Console.log (5+'5');

Answer: 55

14. Console.log (5-'2');

Answer: 3

15. Console.log (true + true);

Answer: 2

16. console.log ([] == false);

Answer: true

17. Console.log (0 || "TamilLoft");

Answer: TamilLoft

18. Console.log (null ?? 'Guest');

Answer: Guest

19. let a = 5; console.log (a>3 ? 'Yes' : 'No');

Answer: Yes

20. for (let i=1; i <=3; i++) { console.log(i); }

Answer: 1
2
3

Section - C

Coding Problems with Input & Expected Output

21. Reverse a String (without Using reverse())

Input:

TamilLoft

Expected Output:

tfoLlimaT

Answer:

```
let str = "TamilLoft";
let reverse = "";
for (let i = str.length - 1; i >= 0; i--) {
}
console.log(reverse);
```

Q2. Check wheather a Number is prime

Input:

17

Output:

17 is a prime Number

Answer:

```
let num = 17;
let prime = true
```

Answer:

```
let arr = [12, 45, 67, 23, 89, 10]
let largest = arr[0];
For ( let i = 1; i < arr. length ; i++) {
    if (arr[i] > largest ) {
        largest = arr [i];
```

25. Count the Number of Vowels in a String

Input:

Javascript

Expected Output

Total Vowels : 3

Answer:

```
let str = "Javascript";
let count = 0;
```

27. Sum of Even Numbers from 1 to 100

Input: 1 to 100

Expected Output: 2550

Answer:

```
let sum=0;
for (let i=2; i<=100; i+=2){
  sum += i;
}
console.log(" Total Sum :", sum);
```

28. Create a factorial function:

Input : 5

Expected Output: 120

Answer:

```
function factorial (n){
   let fact = 1;
   for ( let i=1; i <= n; i++ ){
      fact *= i;
   }
   return fact;
}
console.log("factorial :", factorial(5));
```

29. print Star Pattern:

input
Rows = 5

Expected Output.

```
*
* *
* * *
* * * *
* * * * *
```

Answer:

```
for ( let i = 1; i <= 5; i++ ){
let star = "";
for ( let j = 1; j <= i; j++ ){
}
console. log (star);
}
```

30. Print keys and Values of Student Object

Input:

```
Const student = { name : "Arum", age 20, department : "cse"};
```

Expected Output:

```
name : Arum
age : 20
department : cse
```

Answer:

```
Const Student = {
    name : "Arum",
    age : 20.
    department : "CSE"
};
for (let key in Student) {
  console.log(key + " ; " + Student [ key]);
}
```

31. Sum of Array Using for ... of

Input:

[10, 20, 30, 40, 50]

Expected Output:

150

Answer:

```
let arr = [10, 20, 30, 40, 50];
let sum = 0;
for ( let value of arr) {
    sum += value
}
Console.log ("Sum:", Sum);
```

32. Print Object keys Using for

input: const employee = { id = 101, name : "Rahul", role : Developer'};

Expected Output: id
name
role

Answer:

```
const employee = {
    id = 101,
    name : "Rahul",
    role : "Developer"
};
for ( let key in employee){
console.log (key);
}
```

33. Login Validation

Input: Username : admin
password : admin 123

Expected Output: Login Successfully

Answer:

```
let Username = "admin";
let password = "admin 123";
if (username === "admin" && password === "admin 123")
{
   console.log ("login Succesful");
} else {
   console.log ("Invalid Login");
}
```

34. Optional Changing

Input: Const User = { name: " John" };

Expected Output: undefined

Answer:

```
const user = {
    name: "john"
};
console.log (user. address?.city);
```

35. Student Grade Calculator:

Input:

Tamil: 90

English: 80

maths: 95

Science: 75

social: 88

Expected Output:

Total : 428

Average 85.6

Grade : A

Answer:

```
let tamil = 90;
let english = 80;
let maths = 95;
let science = 75;
let social = 88;
let total = tamil + English + maths + science +
                                          social;
let average = total / 5 ;
let grade;
if ( average >= 85){
   grade = "A";
} else if (average >= 70){
   grade = "B";
} else if (average >= 50){
   grade = "c";
} else {
      grade = "fail";
}
console.log ( "Total :", total);
console.log (" Average:", average);
console.log (" Grade :", grade);
```